OTA Compliance

Price discovery, monitoring, and review

compliance.osakiplus.com

I built OTA Compliance to make advertised-price review a repeatable process. The application finds relevant product listings, checks prices against minimum advertised price (MAP) references, and brings the latest results and source context into one dashboard. Its scope is the selected product models, configured dealer websites, and additional pages found through external search.

Keeping up with changing listings

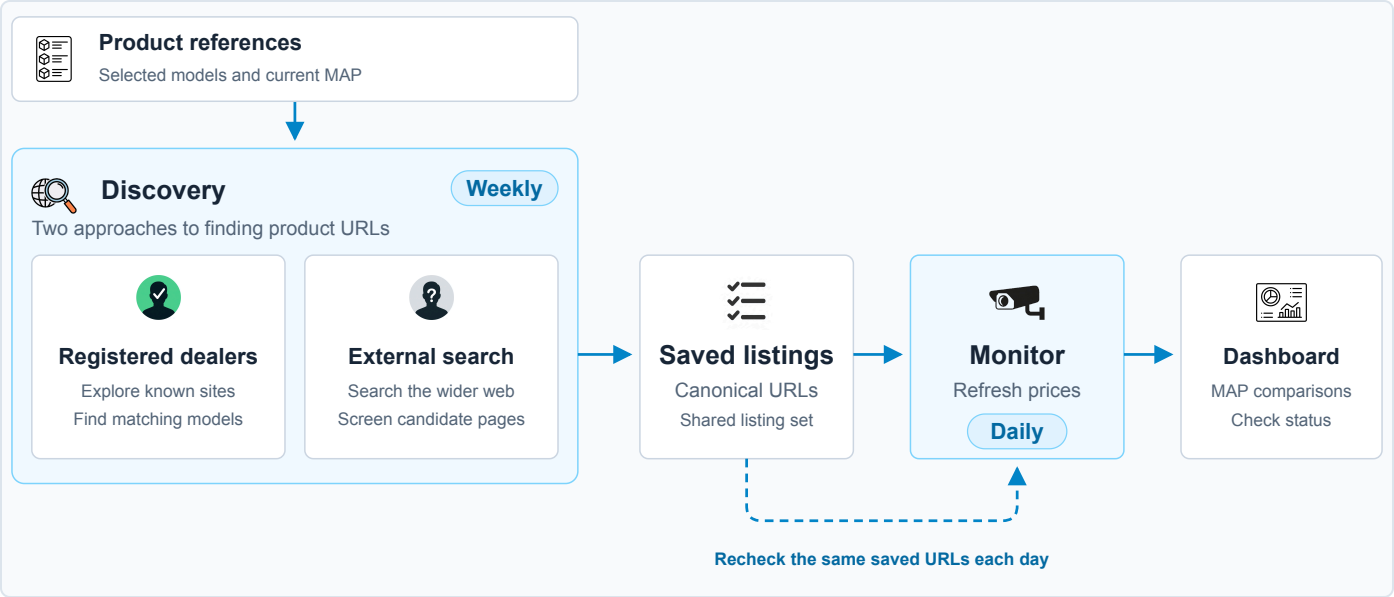
Reviewing advertised prices by hand means finding the right product page, identifying the model, comparing the price with the current MAP, and keeping enough context to revisit it later. That work repeats across websites and over time. Product aliases, different page layouts, and changing URLs make the comparison harder to maintain consistently.

A useful result also needs a source and a capture time. A page may change, become unavailable, or block a request. I designed the project to keep those outcomes visible alongside successful observations, so reviewers can see both the available pricing evidence and the gaps that still need follow-up.

Finding pages and checking them again

The workflow separates finding product pages from rechecking their prices. Discovery contains two approaches, registered dealer discovery and external search, which feed one saved listing set.

Discovery expands the listing set weekly; Monitor refreshes saved price observations daily.



Registered dealer discovery and external search share one listing set for daily monitoring.

Two different discovery problems

Registered dealer discovery addresses coverage on known websites. The system knows which dealers to inspect, but still has to find the right product pages among different catalogs and page structures. It uses available feeds, collections, site searches, and sitemaps to locate the selected models within those sites.

External search looks for additional listings across the wider web, including sites outside the registered dealer list. It uses Serper to retrieve structured Google Search and Shopping results for product names and aliases. Returned titles and URLs are screened for unrelated content, ambiguous model matches, and duplicates before candidate pages are checked.

Separating the approaches lets each use the right search method and request limits.

Monitor: daily price checks

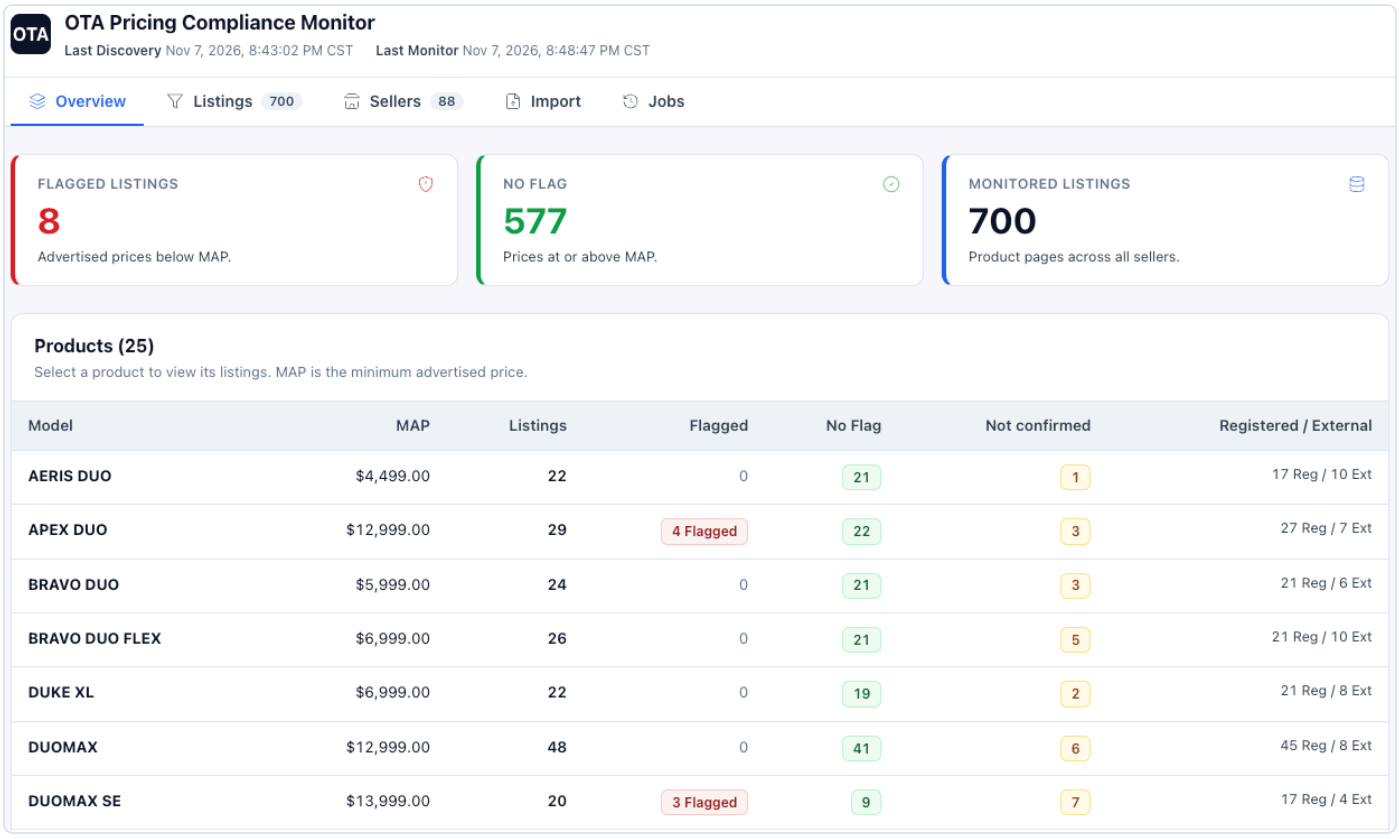
Monitor runs daily against eligible saved URLs. It confirms the product model and extracts the advertised price, currency, and availability from product pages or public catalog data. Each attempt saves its capture time and supporting evidence while preserving earlier checks.

If a page rejects access or its price cannot be confirmed, the website outcome and available source context remain visible. These listings stay unresolved for review.

Reviewing the latest results

The dashboard groups listings by model and seller and shows each latest check's price, MAP comparison, flag, and capture time. Listing details provide the check status and available source context.

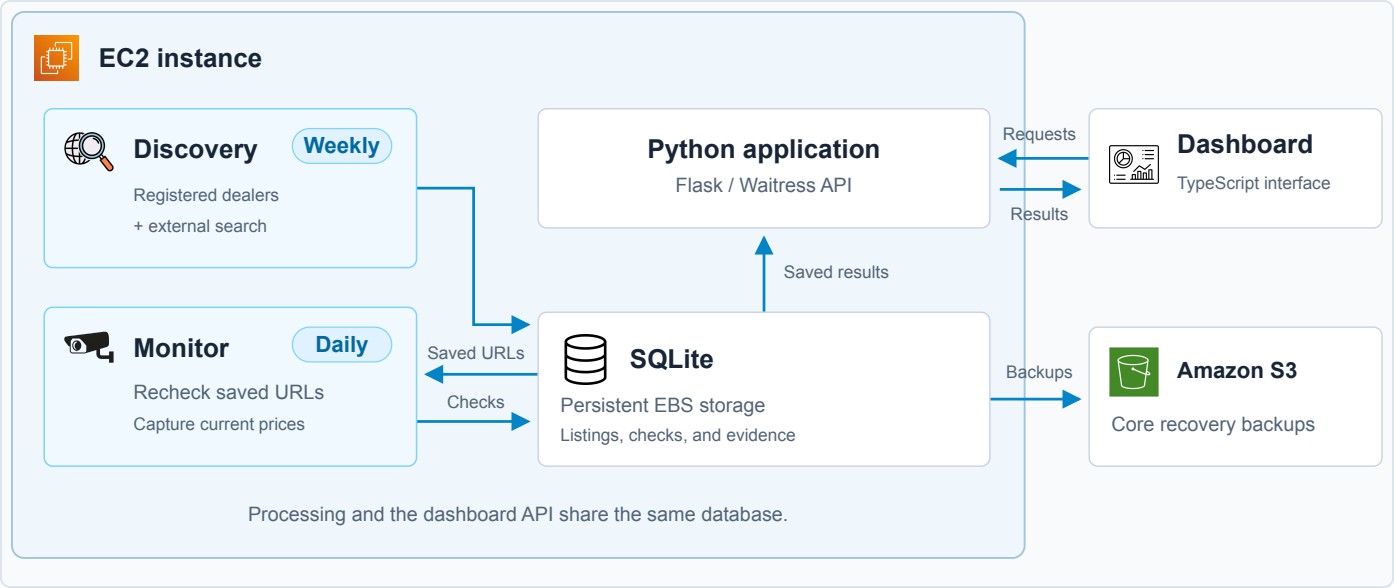
Earlier checks remain stored in SQLite with their original timestamps, MAP values, and captured facts.



The dashboard summarizes monitored listings and highlights product models with flagged prices or checks that still need confirmation.

Running the service

The TypeScript dashboard is backed by a Python application using Flask and Waitress on an EC2 instance. Discovery and Monitor store their results in SQLite on persistent EBS storage, alongside captured evidence and run summaries. A private S3 bucket stores recovery bundles for the database and core configuration.



Persistent EBS storage keeps operational data; S3 holds core recovery backups.